XRPL Validator — Daily Engineering Report · 2026-10-10

Generated 2026-10-11 00:21 EDT. Scope: XRPL work only — the git history of xrpl-validator and the XRPL Rust SDK, the live native-engine shadow, the reference rippled node, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **4 commits** (xrpl-validator 4)
- **2 engine findings fixed** (F418–F419)
- Live shadow: 21,891/21,893 ledgers fully matched, 1,863,905 transactions, 2 engine receipts in the current soak
- Amendment **fixCleanup3_4_0** held majority (since 2026-10-09 09:34 UTC); activation due ~2026-10-23 09:34 UTC

Live validator and native engine

Snapshot at generation time.

Measure	Value
Ledgers compared (current run)	21,893
Ledgers fully matched	21,891
Transactions applied natively	1,863,905
Result mismatches	0
Byte mismatches	5
Missing / extra keys	0 / 0
Batch inner-result mismatches	0
Engine receipts (current soak)	2
Live engine build running since	2026-10-10 00:44 EDT
Reference rippled	3.4.1, full, validated #107,576,027

Engine findings fixed

- **fix(ledger): F418 — a permission by name and by number is one permission, so re-filing a delegation changes nothing (Mainnet #107570369 EE825DEED447)** (`452f251`, xrpl-validator) A DelegateSet re-filed the two permissions its Delegate object already held (OfferCreate, OfferCancel). Mainnet's metadata has only the sender's AccountRoot: rippled's ApplyStateTable never emits a node whose final content equals its original. The engine wrote the object back with the transaction's PermissionValue names, where the decoded original holds their numbers (8, 9), so the write-back read as material, was …
- **fix(ledger): F419 — AMMCreate links the AMM, sends the LP tokens, then Amount, then Amount2, as rippled does (Mainnet #107572028 AC9F1F115224)** (`2053735`, xrpl-validator) An issuer pooled two of its own tokens (PGold / PCopperLockpick), so all three new trust lines went into its own owner directory, whose last page

(0x10) had one slot left. rippled's applyCreate sends the LP tokens before the assets: the LP line took the slot and both asset lines opened page 0x11. Ours funded the pool first and minted last, so the LP line landed on 0x11 and the pages' contents and two HighNodes …

Engineering changes

- **fix(probe): an RPC error that does not settle a lookup is a failed fetch, not "absent"** (`6bd1612`, xrpl-validator) #107541732 (Payment D8D77D7AB213, PLX -> mXRP -> FARM through two AMM pools): s2 answered the PLX/mXRP pool's ledger_entry {amm} with an error in three runs. rpc() returned the error as data, load_book_pair read it as "no pool", and the replay reported a phantom tecPATH_DRY with a verdict (exit 1). One DX_RPCLOG re-run matched 108/108. RPC_FAILED counted only network failures, so a well-formed JSON error slipped …

Docs and operations

- **docs: fixBatchV1_2 is ported (cycle 168), so task 1.5 is done and the known-gap list starts empty** (`4bb4877`, xrpl-validator)

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-09 09:34 UTC	2026-10-23 09:34 UTC

Amendments enabled on mainnet: 97.

Appendix — commits

Time	Repo	Commit	Subject
01:03	xrpl-validator	`4bb4877`	docs: fixBatchV1_2 is ported (cycle 168), so task 1.5 is done and the known-gap list starts empty
19:00	xrpl-validator	`452f251`	fix(ledger): F418 — a permission by name and by number is one permission, so re-filing a delegation changes nothing (Mainnet #107570369 EE825DEED447)
19:00	xrpl-validator	`6bd1612`	fix(probe): an RPC error that does not settle a lookup is a failed fetch, not "absent"
20:13	xrpl-validator	`2053735`	fix(ledger): F419 — AMMCreate links the AMM, sends the LP tokens, then Amount, then Amount2, as rippled does (Mainnet #107572028 AC9F1F115224)